

A Systematic Review of Body Fat Distribution and Mortality in Older People



We conducted a systematic review investigating body fat distribution in older adults and its association with morbidity and mortality. Our search yielded 2,702 citations. Following three levels of screening, 25 studies were selected to evaluate the association between body fat distribution and comorbidity, and 17 studies were used in the mortality analysis. Most of the selected studies in our analyses used anthropometric measures, e.g., body mass index (BMI), waist circumference, and waist-hip ratio; relatively few studies used direct measures, such as body fat/lean mass, and percentage body fat. Studies reported inconsistent findings regarding the strongest predictor(s) of morbidity and mortality. However, the majority of studies suggested that BMI per se was not the most appropriate predictor of morbidity and mortality in the elderly because of its inability to discern or detect age-related body fat redistribution. In addition, studies using BMI found that the optimal BMI range for the lowest mortality in the elderly was overweight ($25 \text{ kg/m}^2 \leq \text{BMI} < 30 \text{ kg/m}^2$) or mildly obese ($30 \text{ kg/m}^2 \leq \text{BMI} < 35 \text{ kg/m}^2$). Our findings suggest that the current clinical guidelines, recommending that overweight and obesity are major risk factors for increased morbidity and mortality are not applicable to this population. Therefore, the central message of this review is to admonish the government to establish new guidelines specifically for this population, using a combination of body fat distribution measurements, and to certify that these guidelines will not be applied to inappropriate populations.